

Brandon Pardi

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Education

University of California, Davis

PhD in Computer Science (in progress)

September 2024 – June 2029

GPA 4.0

University of California, Merced

BS in Computer Science and Engineering (High Honors)

August 2021 – December 2023

GPA: 3.95

Sierra College

AS in Computer Science, Mathematics, Physics, and Natural Science

August 2017 – May 2021

GPA: 3.35

Fellowships/Honorships

- **US Census Bureau Data Science Fellow** (Summer 2024)
- **Outstanding Student Award for Computer Science and Engineering** (Fall 2023)
- **NSF-CREST Fellow** (Fall 2022 – Spring 2023)
- **Chancellor's Honor List** (Spring 2022)
- **Dean's Honor List** (Fall 2021, Fall 2022, Fall 2023)

Experience

Graduate Student Researcher, Roncali Lab, UC Davis

October 2024 – Present, 30hrs/wk

- Studying the efficacy of performance improvements of track-wise Monte Carlo optical particle physics simulations using Wasserstein Generative Adversarial Networks w/ gradient penalty.
- Applying hyperparameter optimization techniques such as Bayesian optimization with radial basis surrogates, adaptive successive halving, and gradient based subset selection.
- Investigating various probability density function estimation methods, including Gaussian/Epanechnikov Kernel Density Estimation, Bayesian Block Histograms, Gaussian Mixtures, Orthogonal Series, Wavelets, and more.
- Contributing to open source development of GATE v10, a Python-based wrapper for the C++ particle physics simulator Geant4 from Cern labs.

United States Census Bureau Data Science Fellowship

June 2024 – August 2024

- Developed a data processing pipeline for classifying benefits from documents using NLP techniques and LLMs.

Computational Tool Developer, MECHANOB[i]OLOGY Lab at UC Merced

June 2022 – June 2024, 20hrs/wk

- Expedited data analysis workflow.
- Provided intuitive ways for researchers to navigate software.

Lawrence Livermore National Laboratory Data Science Challenge

July 2023, 40hrs/wk for 2 weeks

- Classified different heart diseases from 12 lead ECG data.
- Designed a model architecture with skip connections to predict activation graphs of 75 different points in the myocardium with an R2 score of 0.96.

Project Lead: Automated Rodent Detection Pipeline for Agrecom

January 2023 – May 2023, 20hrs/wk

- Assigned tasks for developing/integrating different facets of the program.
- Trained a custom image classifier to detect rodent presence from cameras.

Skills

Programming Languages

Soft Skills

- Python (Advanced)
- SQL (Proficient)
- C/C++ (Proficient)
- HTML/CSS/JS (Proficient)
- Self-motivated
- Independent
- Team-working
- Adaptability

Technical Skills

- Data Science/Machine Learning
- Temporal & Spatial Modeling
- Mathematics
- Classification, Regression & Generative Models
- Exploratory Data Analysis (EDA)
- Computer Vision
- Linux/Unix
- Software Development
- Algorithm Design
- Database Systems

Domain Expertise

- Biophysics
- Biomedical
- Particle Physics
- Statistics
- Surface Sciences
- Cell Mechanics
- Cardiology
- Audio
- Educational Technology
- Civics and Public Administration

Research Projects

pyQCM-BraTaDio

- Designed Python algorithms to scrub, reduce, analyze, and model thousands of data sets for Quartz Crystal Microbalance with Dissipation experimental data.
- Created graphical user interfaces for intuitive use by researchers.

Marker Tracker

- Employed numerous computer vision techniques to track cell movements and surface areas.
- Created intuitive methods for visualizing all tracked data.

Student-Centric Team Formation for UC Merced Computer Science Capstone Classes

- Designed an algorithm to form teams and pair them with industry projects based on various constraints such as student skill sets and preferences.
- Performed text data mining using a QA LLM to extract skill sets required for projects.

Multi-Model Noise Cancellation Pipeline

- Developed a pipeline with audio classifier to LLM to source separator.
- Used historical context and RLHF to respond to user preferences.

Neural Network Visualizers

- Developed basic neural networks from scratch using no machine learning libraries.
- Created interactive plotting tools to visualize the network's learning process.

Publications

- Pardi BM, Ahmed ST, Flores SJ, Maers LE, Yanez Soto B, Andresen Eguiluz RC. *pyQCM-BraTaDio: A tool for visualization, data mining, and modelling of Quartz crystal microbalance with dissipation data*. Journal of Open Source Software, 9(99), 6831 (2024). DOI: 10.21105/joss.06831
- Smith AM, Inocencio DG, Pardi BM, Gopinath A, Andresen Eguiluz RC. *Facile determination of the Poisson's ratio and Young's modulus of polyacrylamide gels and polydimethylsiloxane*. ACS Applied Polymer Materials 6(12): 2405-2416 (2024). DOI: 10.1021/acsapm.3c03154
- Pardi BM, Adhikari A, Chandrasekhar S. *Student Centric Team Formation: Inspiring Project Success Through First Come First Serve Skill/Preference Scoring*. 2025 (in preparation)

Contactable References Available Upon Request